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Optimizing boundary dynamics for nested named entity recognition via semantic refinement and trimming

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ABSTRACT

Nested Named Entity Recognition (Nested NER) addresses the complex task of identifying and classifying entity spans embedded within other entities in textual data. Despite advances, existing span-based methods primarily rely on exhaustive span enumeration without adequately accounting for subtle semantic differences at entity boundaries, leading to boundary ambiguity and inaccurate entity delineation. This limitation significantly affects the precision of nested entity detection, particularly in densely nested contexts. To overcome these challenges, we propose a novel approach named Semantic Refinement and Trimming (SRT). Specifically, SRT employs a biaffine attention mechanism to construct detailed semantic representations for candidate spans, which are subsequently refined through a Boundary-aware Semantic Refinement Module (BSRM). This module leverages a learnable convolutional kernel to explicitly capture fine-grained semantic differences between overlapping spans, addressing boundary ambiguity at the semantic level. Moreover, we introduce a Boundary Trimming Module (BTM), which effectively reduces irrelevant span noise through a dual-pathway architecture, simultaneously facilitating top-down semantic refinement and bottom-up semantic restoration. Extensive experiments conducted on established nested (ACE04, ACE05, GENIA) and flat (CoNLL03) NER benchmarks demonstrate that our proposed SRT method achieves state-of-the-art performance.

1. Introduction

Named Entity Recognition (NER) involves computational techniques aimed at identifying and categorizing entities within textual data, assigning them semantic types such as *person* (PER), *location* (LOC), and *geo-political entity* (GPE) (Lample et al., 2016). As shown in Fig. 1, the span “Reporter Cael Dinno of Britain’s ITN” is labeled as PER, “Alabama” and “Florida” as GPE, and “the Alabama-Florida line” as LOC. As a core NLP task, NER not only facilitates the extraction of structured information from unstructured text but also serves as the foundation for a wide array of downstream applications: relation extraction leverages NER outputs to link entities via semantic relations (Luo et al., 2024; Tang et al., 2022), event extraction identifies and classifies event triggers and arguments in context (Fei et al., 2022; Liu et al., 2022), and knowledge graph reasoning integrates entity mentions into graph structures for inference and question answering (Dong et al., 2024; Liang et al., 2023). The precision and recall of NER systems thus directly influence the effectiveness of these higher-level tasks, underscoring the need for

models that can handle increasingly complex linguistic phenomena with high accuracy and robustness.

Historically, the research and development of NER systems have primarily concentrated on identifying flat, or non-nested, entities (Chiu & Nichols, 2016; Lample et al., 2016; Ma & Hovy, 2016). This focus largely arises from conventional annotation practices that have typically favored simpler, flat entity structures, alongside the widespread adoption of sequence-labeling models such as Conditional Random Fields (CRF) and Long Short-Term Memory networks (LSTMs) (Finkel & Manning, 2009). Despite these methodological preferences, natural language frequently contains nested entity structures, wherein entities are embedded within other entities, forming complex hierarchical relationships. Empirical evidence from benchmark datasets, such as ACE04 and ACE05, shows that over 30% of sentences contain nested entities (Katiyar & Cardie, 2018), underscoring a significant gap in current methodologies and emphasizing the need for novel solutions capable of addressing these sophisticated linguistic patterns.

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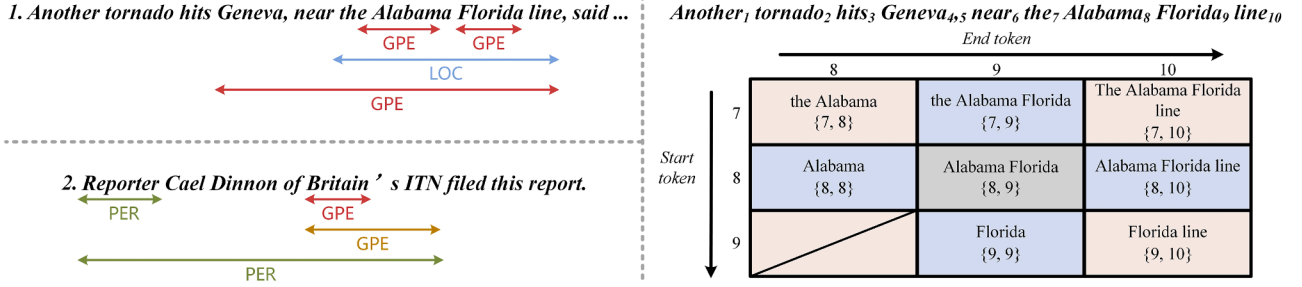


Fig. 1. Examples of nested entities from the ACE Corpus, illustrating boundary ambiguity in nested named entity recognition. The matrix visualizes span interactions on the right, highlighting overlapping spans and shared tokens that contribute to the challenge of distinguishing nested entities.

To tackle the inherent complexities of nested entity structures, several advanced methodological paradigms have emerged. Hypergraph-based methods, for instance, effectively represent nested entities by mapping them onto hypergraph substructures, thus enabling simultaneous processing and classification of overlapping entity spans (Katiyar & Cardie, 2018; Lu & Roth, 2015; Muis & Lu, 2017). Another prominent approach involves hierarchical modeling, which explicitly captures entities at multiple levels of granularity to represent intricate nesting relationships accurately. Although these hierarchical methods offer comprehensive modeling capabilities, they typically incur considerable computational overhead due to the necessity of processing multiple hierarchical paths and layers (Ju et al., 2018; Shibuya & Hovy, 2020; Wang et al., 2020). Conversely, Seq2Seq models address computational efficiency concerns by linearizing the nested structures; however, their sequential processing nature renders them susceptible to cascading errors, particularly in deeply nested or complex scenarios (Straková et al., 2019; Yan et al., 2021; Yang & Tu, 2022; Zhang et al., 2022). This susceptibility underscores the need for methodologies that better capture the nuances of nested entities without incurring significant computational burdens or error propagation.

A representative illustration of the intrinsic complexity and persistent challenges in nested named entity recognition (NER) is evident within benchmark corpora such as ACE05. For instance, consider the sentence from the ACE05 dataset illustrated in Fig. 1: “Another tornado hit Geneva, near the Alabama-Florida line.” Within this brief segment, the entities “Alabama” and “Florida” are each individually annotated as geographically distinct entities (GPE), whereas the composite span “the Alabama-Florida line” is collectively labeled as a location (LOC). This example highlights substantial overlap among entities, with multiple spans sharing head tokens, tail tokens, and internal tokens, thus creating intricate hierarchical structures. Consequently, such token-level overlap introduces significant ambiguity in entity boundary identification, challenging conventional NER models’ capacity to delineate precisely where one entity concludes and another begins.

Traditional span-based NER frameworks often rely heavily upon generalized span encoding techniques, such as pooling boundary token embeddings or concatenating start and end token representations, to derive span-level semantic embeddings. Although these methods have demonstrated empirical effectiveness in simpler, flat-structured scenarios, they exhibit critical limitations when confronted with dense nesting and semantic overlap. Specifically, these encoding strategies tend to produce insufficiently discriminative representations, incapable of adequately capturing subtle yet critical semantic distinctions between overlapping but hierarchically separate entity spans. In the previously mentioned ACE05 example, generic encodings struggle to differentiate between the individual “Alabama” and “Florida” spans versus the encompassing “the Alabama-Florida line” span, thereby exacerbating boundary ambiguity and precipitating erroneous classifications or false-positive entity detections. Moreover, these ambiguities escalate notably under

practical operational conditions, where exhaustive enumeration of candidate spans yields an overwhelming quantity of irrelevant or semantically nonsensical spans, such as “Geneva, near the Alabama-Florida” or “tornado hit Geneva, near”. The proliferation of such lengthy, non-entity spans markedly inflates the complexity of the candidate evaluation process, substantially burdening computational resources and negatively impacting model training and inference efficiency (Tan et al., 2020; Wan et al., 2022). In essence, this indiscriminate inclusion of candidate spans significantly expands the search space, diluting the model’s attention from genuinely relevant entities, ultimately jeopardizing overall accuracy.

To provide deeper insight into these complexities, Fig. 1 (right side) visually demonstrates span-level interactions through a structured representation in the form of a 3×3 span matrix. Within this visualization, the non-entity span “Alabama Florida” centrally occupies the matrix, covering tokens from positions 8 to 9, thus highlighting the intricate relationships between overlapping spans. Horizontally aligned spans within the matrix share common head tokens, vertically aligned spans share tail tokens, and diagonally situated spans represent various degrees of token overlap. This structured visualization underscores the inherent intricacy of entity boundary detection under nested conditions, revealing subtle yet crucial semantic interactions that traditional encoding approaches frequently overlook. Without explicitly tailored semantic guidance mechanisms designed to resolve these nuanced distinctions, conventional NER models remain fundamentally limited, unable to effectively distinguish overlapping spans and consequently leading to compromised accuracy and reliability in nested entity recognition.

Given these inherent challenges in accurately identifying nested entities and addressing boundary ambiguity, we propose a novel approach termed Semantic Refinement and Trimming (SRT). Unlike traditional span-based frameworks, our proposed SRT explicitly integrates mechanisms that emphasize fine-grained semantic differences among overlapping spans, thus enhancing the model’s discriminative capability within densely nested contexts. To explicitly resolve boundary ambiguities arising from token overlap, we further introduce the **Boundary-aware Semantic Refinement Module (BSRM)**. Central to the BSRM is a Boundary-aware Semantic Refinement (BSR) operator, carefully designed to discern subtle yet critical semantic distinctions between overlapping entity spans. Inspired by convolution-based edge-detection techniques commonly used in computer vision, the BSR operator applies a learnable convolutional kernel to span representations, systematically highlighting key semantic regions that differentiate closely related spans. Following convolution, a differentiation phase adaptively adjusts span embeddings according to local semantic context, effectively sharpening distinctions at boundaries where traditional methods commonly struggle. In addition to refining semantic distinctions, practical nested NER scenarios often encounter substantial noise from irrelevant or excessively lengthy candidate spans, substantially increasing computational complexity and impairing classification accuracy. To directly address this issue, SRT includes an innovative **Boundary Trimming Module (BTM)**.

Employing a dual-pathway architecture, the BTM conducts simultaneous top-down semantic refinement and bottom-up detail restoration. The top-down refinement pathway systematically eliminates low-quality candidate spans by modeling high-level semantic interactions among spans, while the bottom-up restoration pathway preserves critical fine-grained details necessary for accurate entity boundary delineation. Through this dual mechanism, BTM effectively trims noisy candidate spans from the prediction space, maintaining only semantically relevant candidates and significantly improving computational efficiency and entity recognition accuracy. Our main contributions are summarized as follows:

- We tackle the persistent boundary ambiguity problem in nested NER by explicitly modeling and emphasizing subtle semantic differences between overlapping spans. This targeted semantic refinement significantly alleviates the boundary insensitivity prevalent in densely nested contexts, enabling accurate differentiation among hierarchically overlapping entities.
- Building on this insight, we introduce an end-to-end framework meticulously designed to discern fine-grained semantic differences at entity boundaries. The framework comprises two integral components, the Boundary-aware Semantic Refinement Module and the Boundary Trimming Module, which collaboratively enhance the precision of entity detection and delineation in complex nested structures.
- Comprehensive experiments on the ACE04, ACE05, GENIA, and CoNLL-E03 datasets demonstrate that SRT outperforms existing state-of-the-art models in both nested and flat NER tasks. Furthermore, detailed ablation studies provide empirical validation of the substantial contributions made by each module within our framework.

2. Related work

In this section, we first survey the evolution of flat Named Entity Recognition (NER) methods, which have laid the groundwork for more complex nested NER tasks. We then move forward to examine the distinct challenges posed by nested entities and review the principal classes of solutions, hypergraph-based, hierarchical, Seq2Seq, and span-based methods, highlighting their respective advantages and limitations.

2.1. Flat named entity recognition

Named Entity Recognition (NER) aims to identify and classify mentions of specific entities within text into predefined categories, such as names of persons, organizations, locations, dates, and other proper nouns. The concept of a "Named Entity" was first introduced at the sixth Message Understanding Conference (MUC-6) (Grishman & Sundheim, 1996). Despite its significance, the diversity and lack of constraints on the types of words that can be named entities complicate the generalization from small data samples (Lample et al., 2016). As a result, traditional approaches have relied heavily on meticulously crafted orthographic features and language-specific resources, such as gazetteers (Mikheev et al., 1999), to enhance performance. However, the development of such resources and features is both time-consuming and expensive.

In recent years, deep learning-based NER models have emerged as the dominant approach, consistently achieving state-of-the-art performance across various benchmarks. Unlike traditional NER methods that heavily depend on manually crafted features and domain-specific knowledge (Krishnan & Manning, 2006; McCallum & Li, 2003), deep learning models excel by automatically discovering hidden patterns and features within the data. Lample et al. (2016) advanced the field by combining bidirectional Long Short-Term Memory (LSTM) networks with Conditional Random Fields (CRF), which allowed for the effective capture of contextual information from both directions in a sentence, thereby improving entity recognition. Additionally, they introduced a

transition-based approach inspired by dependency parsing, using stack LSTMs to incrementally build and label segments of text. This innovation led to the formulation of the NER task as a sequence-labeling problem (Lin et al., 2019b; Liu et al., 2018), which allows for more precise identification of entities within a given context.

The advent of pre-trained language models, such as BERT (Devlin et al., 2019), has further propelled advancements in NER by leveraging large-scale pre-training to develop a deep understanding of context and semantics, leading to significant improvements in model performance. However, despite these advancements, recognizing nested entities (entities that are embedded within other entities) remains a challenging problem. This complexity arises from the intricate structure of nested entities, which requires models to capture multiple levels of context simultaneously (Wang et al., 2022b; Yan et al., 2023b).

2.2. Nested named entity recognition

Nested Named Entity Recognition (NER) is a complex task in Natural Language Processing (NLP) that requires identifying and classifying named entities within text data, where entities can have overlapping and intricate structures. This paper discusses various methodologies developed to tackle this challenge, categorizing them into hypergraph methods, hierarchical methods, Seq2Seq methods, and span-based methods. Each approach is discussed in detail below.

2.2.1. Hypergraph-based methods

The hypergraph-based method, initially proposed by Lu and Roth (Lu & Roth, 2015), maps nested entity structures to sub-graphs within a hypergraph, enabling the classification of these sub-graphs to identify named entities. This approach facilitates the representation of complex and overlapping entities within a unified framework. Muis and Lu (Muis & Lu, 2017) extended this work by introducing a labeling scheme that improves the recognition of overlapping entities by capturing dependencies between entity labels. Katiyar and Cardie (Katiyar & Cardie, 2018) further enhanced the hypergraph model by incorporating a more sophisticated scoring mechanism for sub-graphs, which significantly boosts the model's ability to differentiate between closely related entities. Despite these improvements, hypergraph-based methods can become computationally intensive when handling deeply nested structures, particularly due to the need to handle numerous potential sub-graph formations and their interactions (Fu et al., 2021; Huang et al., 2023).

2.2.2. Hierarchical methods

Hierarchical methods offer a distinct approach by structuring entities into various levels of specificity, enabling a more nuanced and detailed recognition process. Ju et al. (2018) introduced a model that processes entities hierarchically, with each deeper level representing a more specific entity type. This model employs recurrent neural networks (RNNs) to capture dependencies across different levels of the hierarchy, allowing for the accurate recognition of entities with varying levels of nested complexity. Building on this idea, Wang et al. (2020) developed a pyramid sequence labeling framework that uses convolutional neural networks (CNNs) to extract entities from the most specific to the most general. This bottom-up approach effectively manages the complexity of nested entities by leveraging the hierarchical structure to progressively refine entity predictions. The pyramid structure enhances the model's ability to distinguish between closely related entities at different levels of specificity, thus improving overall accuracy.

Further advancements were made by Shibuya and Hovy (Shibuya & Hovy, 2020), who explored suboptimal path decoding to progressively extract entities by considering multiple potential paths through the hierarchy. This method helps mitigate the issue of local optima, which can arise when relying solely on the optimal path for entity extraction. By evaluating multiple paths, the model can more accurately identify entities that may be overlooked by traditional methods. Wang et al. (2021) refined this approach by excluding the influence of the optimal path

to reduce bias. This method, known as path exclusion, ensures that the model does not overly rely on the initially identified optimal path, which can sometimes be suboptimal in complex nested structures. By considering alternative paths and reducing bias towards the first identified path, the model achieves more balanced and accurate entity recognition. However, These models also suffer from high computational costs for nested structures, as each level requires processing dependencies and potential paths through the hierarchy.

2.2.3. Seq2Seq methods

Seq2Seq methods provide a more streamlined, end-to-end approach to NER. These methods typically use models like LSTM-CRF (Straková et al., 2019) to predict the linearized labels, effectively transforming the nested NER task into a simpler sequence labeling problem, or to predict the label for each position in the text (Yan et al., 2021). Zhang et al. (2022) improved this approach by introducing intra-entity and inter-entity de-confounding data augmentation techniques, which help the model generalize better by reducing the bias introduced by entity co-occurrence patterns. Shen et al. (2023b) advanced Seq2Seq methods by designing a dual-slot multi-prompt template, with one slot dedicated to locating entities and the other to typing them. This dual-slot mechanism enhances the model's ability to handle entities in different contexts. Nevertheless, Seq2Seq methods often face difficulties with long-distance dependencies, which can lead to cascading errors in highly nested structures.

2.2.4. Span-based methods

Addressing the limitations of other methods, Sohrab and Miwa (Sohrab & Miwa, 2018) proposed a span-based approach that treats NER as a span prediction problem. This method involves predicting potential entity spans for each token and then filtering and merging these spans to identify nested entities. This foundational approach enables the model to consider each possible span as a candidate entity, significantly improving its ability to capture nested structures. Building on this, Wan et al. (2022) integrated graph structures to better capture relationships between spans. By representing spans as nodes in a graph, this method allows for more sophisticated interactions and dependencies between overlapping spans, enhancing the model's overall accuracy in identifying nested entities. Shen et al. (2021) and Tan et al. (2021) further improved span prediction accuracy by incorporating valuable span patterns. These patterns help the model to recognize common configurations of entity spans, thereby refining its predictions and reducing false positives.

Utilizing the attention mechanism, Yu et al. (2020) introduced a bi-affine attention mechanism to enhance interaction between start and end tokens. This mechanism assigns scores to each span based on the interactions between its boundary tokens, improving the precision of span identification by considering the relationships between span boundaries more effectively. To enhance the model's ability to distinguish between similar spans, Zhang et al. (2023) constructed a span-based contrastive loss function using concatenation to generate span representations. This method focuses on differentiating spans by their contextual features, thus reducing errors in identifying overlapping or similar entities. Shen et al. (2023a) redefined NER by modeling it as an entity boundary-denoising diffusion process. This innovative approach refines and clarifies noisy spans to produce accurate named entities by progressively denoising the span boundaries, which significantly enhances the robustness of the model against noisy or ambiguous input data.

Despite these advancements, span-based models often struggle to capture subtle semantic differences between highly overlapping spans, frequently generating semantically similar representations for these spans. This issue primarily arises because techniques such as pooling (Eberts & Ulges, 2020; Li et al., 2021; Shen et al., 2021), concatenation (Li et al., 2021; Tan et al., 2020; Zheng et al., 2023), or integration (Shen et al., 2023a; Yuan et al., 2022; Zhu & Li, 2022) are typically applied

to the start and end positions of an entity. These techniques focus predominantly on the boundary tokens, often neglecting the rich internal structure and contextual information within the span itself, leading to boundary insensitivity. This boundary insensitivity makes it difficult for models to effectively distinguish between entities with overlapping or closely situated boundaries. For instance, pooling methods aggregate features from the start and end tokens of a span, but may overlook critical context within the span, leading to similar representations for different spans. Similarly, concatenation approaches, while combining features from the boundary tokens, fail to account for the internal structure of the span, limiting the model's ability to differentiate between overlapping entities.

To address the challenge of boundary insensitivity in nested Named Entity Recognition, we propose explicitly incorporating span semantic difference features into the task. By introducing these features, the model can effectively capture the subtle semantic distinctions between entity and non-entity spans, enabling a more precise understanding of where entities begin and end. This approach allows the model to learn more robust and refined span representations, enhancing its ability to differentiate between overlapping or nested entities and improving the overall accuracy of boundary detection in complex NER scenarios.

3. Proposed methodology

This section presents the architecture of our approach, as outlined in Fig. 2. At its core, our framework re-conceptualizes nested named entity recognition (NER) through the lens of boundary identification and filtration.

Nested NER as Boundary Detection. The nested NER task involves parsing an input sequence $X = \{x_1, x_2, \dots, x_n\}$ to both pinpoint and categorize entities into a predefined set of types $T = \{t_1, t_2, \dots, t_k\}$. An entity is characterized as a triplet (s_i, e_i, t_i) , with s_i and e_i indicating the start and end positions of the entity span and $t_i \in T$ representing its assigned type. This formulation ensures accurate identification of entity boundaries in the sentence. For any sequence of n tokens, the number of possible spans is given by $n(n+1)/2$.

3.1. Span semantic encoder

Given an input sequence $X = \{x_1, x_2, \dots, x_n\}$, we utilize a pre-trained BERT model (Devlin et al., 2019) to encode each token x_i into a dense vector representation. This step generates token-level embeddings $\mathbf{H}_{enc} = \{\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_n \mid \mathbf{h}_i \in \mathcal{R}^{h \times 1}\}$, where h corresponds to the embedding dimension, and n is the number of tokens.

Next, two separate feed-forward networks transform these token embeddings into semantic representations for the start and end positions of each span, yielding $\mathbf{h}_s, \mathbf{h}_e \in \mathcal{R}^{l \times h}$. Here, l represents the sequence length. To capture the structural characteristics of spans, we integrate the start and end embeddings with span width information using a biaffine transformation. The span width is encoded into $\mathbf{w}_{ij} \in \mathcal{R}^{c \times 1}$, allowing us to construct a span representation tensor $\mathbf{M}^0 \in \mathcal{R}^{l \times l \times f}$, where c is the dimensionality of the width embedding, and f represents the feature size produced by the biaffine layer.

For a span S_{ij} extending from token i to token j , its span vector $\mathbf{M}_{ij}^0$ is determined as:

$$\mathbf{h}_s = \text{GELU}(\mathbf{H}_{enc} \mathbf{W}_s); \quad \mathbf{h}_e = \text{GELU}(\mathbf{H}_{enc} \mathbf{W}_e), \quad (1)$$

$$\mathbf{M}_{ij}^0 = [\mathbf{h}_s[i] \parallel \mathbf{h}_e[j] \parallel \mathbf{w}_{ij}] \mathbf{W} + \mathbf{h}_s[i] \mathbf{U} \mathbf{h}_e[j]^T,$$

where $\mathbf{W}_s, \mathbf{W}_e \in \mathcal{R}^{h \times h}$, $\mathbf{W} \in \mathcal{R}^{(2h+c) \times r}$, and $\mathbf{U} \in \mathcal{R}^{h \times r \times h}$ are learnable parameters. r denotes the feature size of the biaffine model, and $\parallel$ denotes concatenation. GELU introduces non-linearity into the token representations. Notably, when $\mathbf{M}_{ij}^0$ lies off the main diagonal of the matrix $\mathbf{M}^0$, the span representation $\mathbf{S}_{ij}$ manifests in two symmetrical configurations along the diagonal. This symmetry enables the model to consider both forward and backward contexts, ensuring a comprehensive view of overlapping spans and their subtle differences.

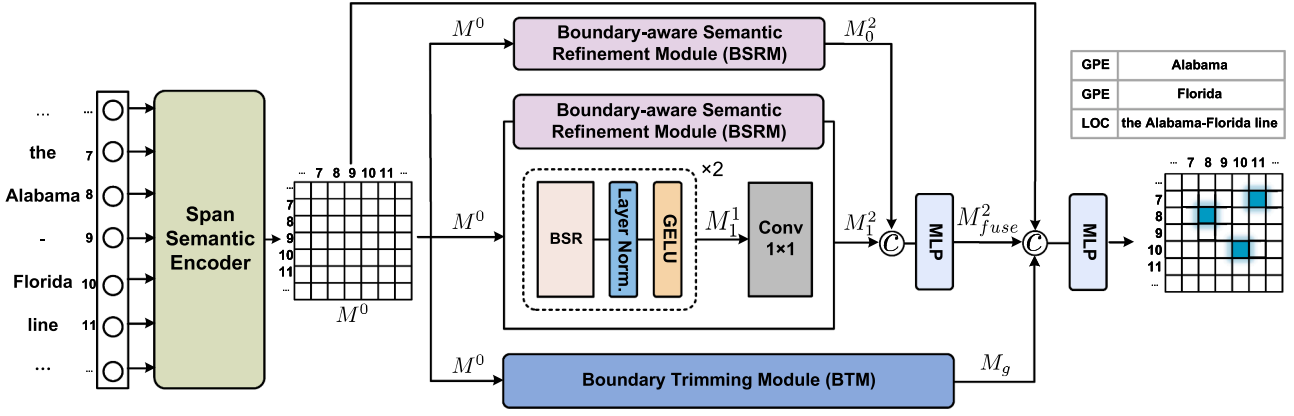


Fig. 2. Architecture of SRT. Given an input token sequence, the Span Semantic Encoder constructs an initial biaffine span representation matrix M^0 . This matrix is then processed in parallel by the Boundary-aware Semantic Refinement Module (BSRM) and the Boundary Trimming Module (BTM). The BSRM applies two layers of the novel Boundary-aware Semantic Refinement (BSR) operator, each comprising a learnable convolutional kernel, LayerNorm, and GELU to highlight fine-grained semantic differences and outputs a fused refined matrix M_{fuse}^2 . $\odot$ denoted the concatenate operation. LN is LayerNorm Layer (Ba et al., 2016). Details of Boundary-aware Semantic Refinement operator and Boundary Trimming Module are shown in Fig. 3.

3.2. Boundary-aware semantic refinement module

To accurately capture nuanced semantic distinctions between spans, we propose the **Boundary-aware Semantic Refinement** (BSR) operator, drawing inspiration from established computer vision techniques, including the Roberts cross operator (Roberts & Lawrence, 1965). By adapting traditional gradient operations to the local semantic context of each span, the BSR operator enhances the detection of subtle differences between overlapping spans. Its functionality is divided into two primary stages:

1) *Masking Phase.* A learnable convolutional kernel evaluates local semantic regions, generating a mask matrix mask_{x_0} that highlights critical spans for semantic differentiation. The convolutional process focuses on the region R surrounding the center position x_0 in the span representation matrix M . The $\arg \max$ operation identifies the position within R , excluding x_0 , with the highest semantic score, thus defining a mask that emphasizes the most relevant spans:

$$I_{x_0} = \underset{i \in R \setminus \{x_0\}}{\text{Argmax}} [LN(Conv(M^0))]_{i,x_0}, \quad \text{mask}_{x_0}[i] = \begin{cases} 1 & \text{if } i = I_{x_0} \text{ or } i = x_0 \\ 0 & \text{others} \end{cases} \quad (2)$$

Where $i \in R \setminus \{x_0\}$ denotes i ranges over all elements of the set R , except for the particular element x_0 . Specifically, consider the example on the right hand side of Fig. 1. We set the end index 9 and examine the spans $(i, 9)$ for $i \in \{7, 8, 9\}$. After applying the learnable convolution and layer normalization, we obtain scores for "the Alabama Florida" at position (7, 9), "Alabama Florida" at (8, 9), and "Florida" at (9, 9). Excluding the central span x_0 (8, 9), we select the highest score neighbor (7, 9) ("the Alabama Florida"), the construct the mask and zeroing out all other positions.

2) *Differentiation Phase.* Span representations are refined by element-wise multiplication of the mask matrix with a predefined weight matrix w_f . This enables adaptive adjustments to span representations based on the local context:

$$\text{BSR}(x_0) = \sum_{x_n \in R} \text{mask}_{x_0} \cdot w_f \cdot M_{x_n}^0, \quad (3)$$

where $M_{x_n}^0$ is the span semantic matrix at position x_n , and w_f is a fixed weight matrix.

The BSR operator is fundamental to the BSRM, with each layer incorporating two BSR blocks to model both first-order and second-order

semantic variations between spans:

$$\text{BSRBlock}(\ast) = \text{GELU}(\text{LN}(\text{BSR}(\ast))), \quad (4)$$

$$M_{lr}^1 = \text{BSRBlock}(\text{BSRBlock}(M^0)),$$

where $lr \in \{0, 1, \dots, N\}$, and $N + 1$ denotes the total number of layers. Note that the residual connections within the BSR blocks are not explicitly shown here for simplicity.

To facilitate effective training of the BSRM, which incorporates a non-differentiable Argmax operation, the Gumbel-softmax estimator (Jang et al., 2016) is applied. Additionally, to ensure consistency in semantic differentiation, the two BSR operations use fixed weight matrices with opposing signs. Specifically, the matrix w_f is used for the first BSR operation, and its counterpart w'_f , containing negated elements, is used for the second:

$$w_f = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & -1 \end{bmatrix}, \quad w'_f = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad (5)$$

Following these refinements, the BSRM integrates the resulting semantic features into a unified span boundary matrix M_{fuse}^2 using a linear layer:

$$M_{lr}^2 = \text{Conv}_{1 \times 1}(M_{lr}^1), \quad M_{fuse}^2 = W_{fuse} \underbrace{(M_0^2 \parallel \dots \parallel M_{lr}^2)}_{N+1} + B_{fuse}, \quad (6)$$

where W_{fuse} and B_{fuse} are learnable weights and biases, $\text{Conv}_{1 \times 1}$ denotes a 2D convolution with a 1×1 kernel, and $\parallel$ indicates concatenation.

3.3. Boundary trimming module

To mitigate the interference of spurious or low-quality candidates, which often obscure true entity boundaries and inflate false-positive rates, we propose the **Boundary Trimming Module** (BTM). The BTM operates on the initial span representation matrix and systematically excises irrelevant spans, thereby sharpening the focus on genuine entity boundaries. Architecturally, the BTM adopts a dual-pathway design: a top-down pathway of consecutive convolutional blocks abstracts high-level semantic interactions among spans, while a complementary bottom-up pathway up-samples these abstractions to restore fine-grained boundary details. Lateral connections between corresponding levels of the two pathways ensure that restored details remain contextually aligned with the abstracted features. We describe the procedures as follows:

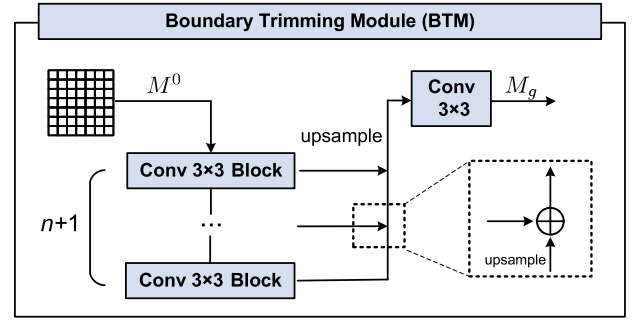
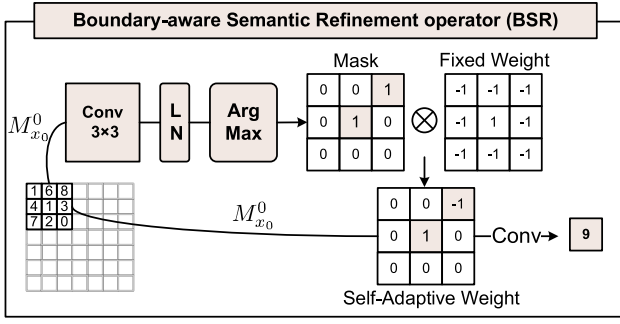


Fig. 3. Detailed visualization of the two core SRT components. The Boundary-aware Semantic Refinement (BSR) operator refines each span representation $M^0_{x_0}$ by first applying a 3×3 convolution, LayerNorm, and an Argmax to generate a binary mask, which is then combine with a fixed weight kernel to produce a self-adaptive weight matrix. This weighted span patch undergoes a further 3×3 convolution to yield the refined semantic feature. The Boundary Trimming Module (BTM) comprises $n + 1$ stacked 3×3 convolution blocks that progressively abstract span interactions. Simultaneously, a bottom-up-sampling pathway reinstates fine-grained details via lateral connections. The final convolution outputs the pruned span matrix M_g , which contains the most salient candidate spans for downstream classification.

Top-Down Pathway. This pathway utilizes a series of convolutional blocks that progressively refine the span features. Each convolutional block applies Layer Normalization and GELU activation to efficiently process the input data:

$\text{ConvBlock}(*) = \text{GELU}(\text{LN}(\text{Conv}(**)))$;

$$\mathbf{M}_g^0 = \text{ConvBlock}(\mathbf{M}^0); \quad (7)$$

$$\mathbf{M}_g^i = \text{ConvBlock}(\mathbf{M}_g^{i-1}),$$

where $i \in \{1, 2, \dots, n\}$, and $\mathbf{M}_g^i$ represents the progressively refined features at the i th level.

Bottom-Up Pathway. This pathway focuses on restoring finer details from higher-level features through up-sampling, which preserve relational information such as span dependencies. The up-sampled features are combined with lateral connections to merge information across different layers:

$$\mathbf{M}_g^{i-1'} = \text{upSample}(\mathbf{M}_g^i) + \mathbf{M}_g^{i-1}, \quad (8)$$

$$\mathbf{M}_g = \text{Conv}(\text{upSample}(\mathbf{M}_g^0)).$$

By combining both top-down and bottom-up pathways, the BTM effectively eliminates irrelevant or noisy information and refines the boundaries of entity spans, resulting in clearer and more distinct span representations within the SRT framework.

3.4. Span semantic decoder

To retain the complete semantic information for each span, our model integrates the initial matrix $\mathbf{M}^0$ as a residual component into the fused matrix $\mathbf{M}_{\text{fuse}}^2$. This integration ensures a comprehensive representation of span semantics, which is then processed through a linear decoding layer to produce the prediction logits:

$$\mathbf{p} = \sigma(\mathbf{W}_p(\mathbf{M}^0 \parallel \mathbf{M}_{\text{fuse}}^2 \parallel \mathbf{M}_g) + \mathbf{B}_p), \quad (9)$$

where $\mathbf{p} \in \mathcal{R}^{l \times l \times t}$, $\mathbf{W}_p \in \mathcal{R}^{d \times t}$, and $\mathbf{B}_p \in \mathcal{R}^t$. The variable t denotes the number of entity types, and σ represents the *Sigmoid* activation function. $\mathbf{W}_p$ and $\mathbf{B}_p$ are trainable parameters.

3.5. Training and inference

3.5.0.1. Training. The model is trained using a binary cross-entropy loss function, which minimizes the discrepancy between the predicted labels and the ground truth entity types:

$$\mathcal{L} = - \sum_{0 \leq i, j < l} (y_{ij} \log(\mathbf{p}_{ij}) + (1 - y_{ij}) \log(1 - \mathbf{p}_{ij})), \quad (10)$$

where y_{ij} is the ground truth entity type, l represents the sequence length. The loss for each class is summed to compute the final loss $\mathcal{L}$.

Table 1

Statistics of the datasets used in the experiments. The "Len" column represents the average length of sentences or entities in each dataset.

Datasets		Train	Dev	Test	Len	Overlap rate
ACE04	Sentence	6,297	742	824	23.52	45.68 %
	Entity	22,231	2,514	3,036	2.64	
ACE05	Sentence	7,178	960	1,051	20.59	39.11 %
	Entity	25,300	3,321	3,099	2.40	
GENIA	Sentence	15,023	1,669	1,854	25.41	18.41 %
	Entity	45,144	5,365	5,506	1.97	
CoNLL2003	Sentence	14,041	3,250	3,453	14.38	-
	Entity	23,499	5,942	5,648	1.45	

In line with the SRT architecture, which does not distinguish between the two halves of the matrix during training, errors from both the upper and lower triangular sections of $\mathbf{p}_{ij}$ and $\mathbf{p}_{ji}$ are included, reflecting the symmetric nature of entity representations.

3.5.0.2. Inference. During inference, predictions from symmetrical positions in the matrix are averaged to stabilize the output:

$$\mathbf{p}'_{ij} = (\mathbf{p}_{ij} + \mathbf{p}_{ji})/2. \quad (11)$$

Following Yu et al. (2020), spans that are classified as non-entities are discarded, and the remaining spans are ranked and selected based on the highest predicted probability, ensuring that no overlapping entities are selected.

4. Experiment

This section begins with an in-depth description of the datasets, baseline models, and implementation specifics. Next, we present the evaluation of the SRT model on benchmark datasets, showcasing its effectiveness. To further analyze its capabilities, we conduct a series of ablation studies to examine the contributions of each proposed module. In conclusion, we assess the performance of the suggested modules in handling nested named entities and long entities, supported by relevant use case scenarios to demonstrate the impact of these modules.

4.1. Experimental setup

In this section, we describe in detail the datasets employed in our experiments, outline the suite of baseline models against which SRT is compared, and specify the optimal hyper-parameter settings determined for our proposed framework.

Datasets. We evaluate our model on three commonly used nested NER datasets and one for flat NER, including ACE04,¹ ACE05,² GENIA,³ and CoNLL2003.⁴ ACE04 and ACE05 consist of seven entity categories: Person (PER), Organization (ORG), Geo-Political Entity (GPE), Location (LOC), Facility (FAC), Weapon (WEA), and Vehicle (VEH). The GENIA dataset is composed of five categories: DNA, RNA, Protein, Cell line, and Cell type, while CoNLL2003 includes four types: LOC, MISC, ORG, and PER.

For the ACE datasets, pre-processing was conducted using methods released by Yan et al. (2023a), splitting the data into training, validation, and test sets by an 8:1:1 ratio. For GENIA, we adhered to the classification and data division methods of Li et al. (2022). The CoNLL2003 dataset was processed according to the methodologies of Lin et al. (2019a), Shen et al. (2021), where our model was trained using a combined set of training and development data.

It is crucial to note that Yan et al. (2023a) highlighted a significant observation: despite the utilization of the same dataset in recent studies (Li et al., 2022; Wan et al., 2022; Yuan et al., 2022; Zhu & Li, 2022), variations in pre-processing methods result in differing training dataset statistics. As a result, direct comparisons of model performance across different studies are potentially unfair. To mitigate this issue, we adopted the pre-processing code from Yan et al. (2023a) for our dataset. Subsequently, we re-implemented several baseline models from 2022 and 2023 using the pre-processed dataset and publicly available code. However, due to the unavailability of source code and limited model details, we were unable to fully replicate the Span Graph (Wan et al., 2022), De-bias (Zhang et al., 2022), and ICR (Zheng et al., 2023) models.

Baseline Models. To evaluate the performance of the proposed model, we compare it with the following models on three datasets:

- 1) **SH** (Wang & Lu, 2018): leverage a hypergraph framework to capture intricate dependencies among overlapping entities effectively.
- 2) **DYGIE** (Luan et al., 2019): design to improve information extraction tasks, DYGIE utilizes dynamic span graphs where text spans are represented as nodes, and edges define the relationships between them.
- 3) **Pyramid-Basic** (Wang et al., 2020): introduce a hierarchical approach to address the challenges posed by nested NER, processing entities at varying levels of granularity.
- 4) **Biaffine**: (Yu et al., 2020) leverage a biaffine mechanism to detect nested named entities by predicting the dependency relationships between pairs of words and their corresponding boundaries.
- 5) **Second-Best** (Wang et al., 2021): focus on nested entity recognition by intentionally disregarding the optimal path of the probability graph to improve recognition.
- 6) **Locate-and-Label** (Shen et al., 2021): modeled as a combined task of entity boundary prediction and span classification, leading to enhanced training and inference efficiency.
- 7) **Seq2Seq** (Yan et al., 2021): use a pointer-based approach to convert the entity tagging task into a sequence generation task.
- 8) **Sequence2Set** (Tan et al., 2021): Introduces a novel neural network architecture dedicated to set prediction for nested NER.
- 9) **UIE** (Lu et al., 2022): Presents an integrative framework designed to consolidate various information extraction tasks.
- 10) **Span-Graph** (Wan et al., 2022): Models nested NER via a span-based graph structure, wherein each span is represented as a node and connected by edges to enhance its semantic representation.
- 11) **De-bias** (Zhang et al., 2022): Investigates erroneous biases in the generation process and employs both intra- and inter-entity deconfounding data augmentation techniques to mitigate biases.

12) **BS** (Zhu & Li, 2022): Proposes a boundary smoothing approach that redistributes probability mass from annotated spans to adjacent spans.

13) **Triaffine** (Yuan et al., 2022): Enhances entity recognition performance by capturing diverse interaction information among heterogeneous elements.

14) **W2NER** (Li et al., 2022): Reframes unified NER as a word-word relationship classification problem, thereby avoiding the label conflicts typical of conventional sequence labeling methods.

15) **ICR** (Zheng et al., 2023): Introduces a scale transformation mechanism coupled with a supervised contrastive learning loss to explore both interactive and contrastive relationships among spans.

16) **Bi-TDNN** (Geng et al., 2023): Converts sentences into a planar graph representation, effectively capturing the hierarchical and overlapping structures inherent in nested entities.

17) **BINDER** (Zhang et al., 2023): Frames the NER task as a representation learning problem aimed at maximizing the similarity between the vector representations of entity mentions and their associated types.

18) **DiffusionNER** (Shen et al., 2023a): Employs a diffusion model for the NER task by progressively refining entity boundaries over noisy span representations.

19) **PromptNER** (Shen et al., 2023b): Designs a dual-slot multi-prompt template featuring both a position slot and a type slot to facilitate the localization and classification of entities.

20) **CNNNER** (Yan et al., 2023a): Utilizes convolutional neural networks to model spatial relationships within the score matrix, thereby addressing the challenges of nested named entity recognition.

21) **Parseker** (Wang & Utiyama, 2024): investigates the application of binary representations in structured prediction tasks, such as named entity recognition, by introducing structured contrastive hashing to generate binary labels while preserving structural information.

Hyper-parameters. For the ACE dataset, we employ the pre-trained models RoBERTa-Base (Liu et al., 2019) and BERT-Large (Devlin et al., 2019), each configured with a hidden size of 768. In contrast, for the GENIA dataset, we utilize BioBERT-Base-v1.1 (Lee et al., 2020), which similarly features a hidden layer dimension of 768. Within our BSRM, the total number of layers (denoted as $N + 1$) is uniformly set to 2 across all datasets, and likewise, the BTM is structured with 2 convolution blocks (i.e., $n + 1$) for every dataset. Excluding any supplementary annotations, the convolution operator Conv is implemented with a 3×3 kernel. Moreover, to optimize memory usage, the BSR operator in each BSRM layer shares parameters, differing solely in their fixed weight templates. The configuration of hyper-parameters for the biaffine model adheres to the approach outlined by Yan et al. (2023a), which incorporates a multi-head biaffine attention mechanism. In this setting, we utilize 4 attention heads and introduce a span width embedding of dimension 25. The Gumbel Softmax estimator is employed with its temperature parameter fixed at 1 by default.

Training is performed using the AdamW optimizer (Loshchilov & Hutter, 2019). To mitigate overfitting, gradient clipping is applied to restrict the L2 norm of gradients to a maximum of 5 (Pascanu et al., 2013). A linear warm-up schedule is adopted during the first 10% of the training process to gradually elevate the learning rate, which is then progressively reduced via a linear decay scheduler. Additional hyper-parameters that vary according to the dataset are provided in Table 2.

Table 2
Hyper-parameter settings on different benchmarks.

	ACE04	ACE05	GENIA	CoNLL 2003
Batchsize	48	48	8	8
Epoch	80	80	10	80
Learning rate	2e-5	2e-5	7e-6	2e-5
Biaffine size	120	120	400	120
CNN channel dim	120	120	200	120
Dropout rate	0.2	0.2	0.1	0.2

¹ <https://catalog.ldc.upenn.edu/LDC2005T09>

² <https://catalog.ldc.upenn.edu/LDC2006T0>

³ <http://www.geniaproject.org/genia-corpus>

⁴ <https://www.clips.uantwerpen.be/conll2003/ner/>

Table 3

The performance of various models on the ACE04, ACE05, and GENIA datasets is presented in Table 3. The "Encoder" column lists the pre-trained models used by each method for the ACE datasets, while all models used BioBERT-Base for the GENIA dataset. The symbol † denotes models that were replicated using identical pre-processed data and publicly available code. The top-performing results are indicated in **bold**, while the second-best are shown with underlines. The subscript represents the standard deviation, illustrating the variability of the results (e.g., 88.48₂₃ means 88.48±0.23).

Models	Encoder	ACE04			ACE05			GENIA		
		P	R	F1	P	R	F1	P	R	F1
SH (2018)	GloVe	78.00	72.40	75.10	76.80	72.30	74.50	77.00	73.30	75.10
DYGLIE (2019)	LSTM	–	–	84.70	–	–	82.90	–	–	76.20
Pyramid-Basic (2020)	GloVe	81.10	79.40	80.30	80.00	78.90	79.40	78.60	77.00	77.80
Biaffine (2020)	BERT-base	87.30	86.00	86.70	85.20	85.60	85.40	81.8	79.30	80.50
Second-Best (2021)	BERT-base	86.42	85.71	86.06	83.95	84.67	84.30	79.20	78.16	78.63
Locate-and-Label (2021)	BERT-base	87.44	87.38	87.41	86.09	87.27	86.67	80.19	80.89	80.54
Seq2Seq (2021)	BART-large	87.27	86.41	86.84	83.16	86.38	84.74	78.57	79.30	78.93
Sequence2Set (2021)	BERT-large	88.46	86.10	87.26	87.48	86.63	87.05	82.30	78.70	80.40
UIE (2022)	T5-base	–	–	86.89	–	–	85.78	–	–	–
Span-Graph (2022)	BERT-base	86.70	85.93	86.31	84.37	85.87	85.11	77.92	80.74	79.30
De-bias (2022)	T5-base	86.36	84.54	85.44	83.31	86.56	84.90	81.04	77.21	79.08
BS (2022) †	RoBERTa-base	87.32	86.84	87.08	86.58	87.84	87.20	82.53	78.69	80.56
Triaffine (2022) †	BERT-large	87.13	87.68	87.40	86.70	86.94	86.82	80.42	82.06	81.23
W2NER (2022) †	BERT-large	87.19	87.72	87.45	85.77	87.80	86.76	<u>83.10</u>	79.76	81.39
ICR (2023)	BERT-large	–	–	–	87.11	87.14	87.12	79.02	80.68	79.87
Bi-TDNN (2023)	ALBERT	88.29	87.91	88.10	86.30	88.67	87.47	83.13	79.75	<u>81.41</u>
BINDER (2023) †	BERT-large	87.34	88.30	87.81	87.41	88.34	87.87	81.69	<u>80.85</u>	81.26
DiffusionNER (2023a) †	BERT-large	87.32	87.52	87.42	85.04	<u>88.42</u>	86.70	81.85	79.59	80.70
PromptNER (2023b) †	BERT-large	87.02	88.03	87.52	86.01	88.12	87.05	–	–	–
CNNER (2023a) †	RoBERTa-base	87.33	87.29	87.31	86.70	88.16	87.42	83.19	79.70	81.40
Parserker (2024)	BERT-large	–	–	87.93	–	–	85.90	–	–	78.79
SRT	RoBERTa-base	<u>88.57</u>	88.43	<u>88.45</u> ₁₄	89.16	88.74	88.94 ₃₈	83.01	80.80	81.87 ₁₉
	BERT-large	88.64	<u>88.32</u>	88.48 ₂₃	<u>88.62</u>	88.17	<u>88.39</u> ₃₁			

4.2. Main results

Our evaluation utilizes three primary metrics: *Precision*, *Recall*, and *F1-score* to measure the performance of the models. The assessment adheres to strict evaluation criteria that require exact matches in both entity boundaries and entity categories for a recognition instance to be considered correct. To ensure consistency and robustness in our results, we conducted five independent experiments, each with unique random seed initializations. We then applied rigorous statistical analysis to the aggregate F1-scores derived from these trials. A bootstrap paired t-test at a 5 % significance level was employed to determine the statistical significance of performance differences, ensuring a comprehensive evaluation of the models' consistency across varied initializations.

Table 3 presents a comparative analysis of SRT and various baseline models on the ACE04, ACE05, and GENIA datasets for nested named entity recognition (NER). SRT consistently demonstrates superior performance across all three datasets. The key findings are as follows:

SRT achieves notable improvements in F1-score, with a +1.24 % increase on ACE04 and a +1.75 % gain on ACE05 compared to the previous state-of-the-art models when utilizing RoBERTa-base as the encoder. These improvements highlight SRT's capability in distinguishing entity boundaries under both flat and nested entity scenarios. In addition to improved F1-scores, SRT consistently demonstrates superiority in both Precision and Recall across the ACE04 and ACE05 datasets, showcasing its ability to accurately identify entities while maintaining a low false positive rate.

SRT shows good performance on Biomedical domain, registering a +0.47 % improvement in F1-score compared to the previous SOTA. However, the model shows a slight underperformance in Precision and Recall compared to some existing methods. This discrepancy may be linked to the lower prevalence of nested entities in GENIA (18.41 %) compared to ACE04 (45.68 %) and ACE05 (39.11 %), as indicated in Table 1. We doubt the reduced complexity of GENIA's nested structures could limit the opportunities for SRT's specialized nested NER mecha-

Table 4

Results on CoNLL03 dataset. All Results are evaluated using BERT-Large pre-trained language model. Performance of baseline models are retrieved from their respective original papers.

Models	CoNLL03		
	P	R	F1
W2NER	92.71	93.44	92.07
DiffusionNER	92.99	92.56	92.78
PromptNER	92.48	92.33	92.41
BINDER	93.08	93.57	93.33
SRT	93.84	93.60	93.72

nisms to demonstrate their full capabilities, thereby narrowing the performance gap between SRT and other methods on this dataset.

SRT demonstrates effectiveness in flat NER contexts, we conducted comparisons with four advanced state-of-the-art models on the CoNLL03 dataset: W2NER, DiffusionNER, PromptNER, and BINDER. As illustrated in Table 4, our model demonstrated superior performance, achieving precision, recall, and F1 scores of 93.84 %, 93.60 %, and 93.72 %, respectively. These results underscore the model's robust capability in boundary detection, which significantly enhances its proficiency not only in nested entity recognition but also in the accurate identification of flat entities.

RoBERTa-base outperforms BERT-large in most of our evaluations, despite the latter's larger parameter count. This performance difference may be attributed to RoBERTa's architecture, which focuses more on token-level tasks like masked language modeling (MLM), rather than next sentence prediction (NSP), as is the case with BERT. This emphasis on token-level representation enhances RoBERTa's ability to understand fine-grained semantic distinctions, making it more effective for tasks like nested NER, which require detailed token-level comprehension.

Table 5

Performance on different nested level of entities. All baselines are re-implemented utilizing RoBERTa-base as encoder.

Datasets	Nested Levels	#. of Instances	CNNNER	DiffusionNER	PromptNER	SRT w/o BSRM	w/o BTM	w/o BSRM, BTM	–
ACE04	0	1614	84.49	81.47	82.82	85.54	85.32	85.06	87.44
	1	974	75.11	67.33	68.10	75.67	75.90	75.28	79.88
	2	325	63.35	52.36	53.31	65.46	65.40	62.13	69.17
	3	77	43.82	21.05	16.67	45.73	46.44	45.55	53.17
	4	20	40.82	5.00	18.18	37.21	37.63	37.03	47.65
	≥ 5	26	46.63	11.76	35.29	47.10	50.85	46.65	52.83
ACE05	0	1913	84.48	83.46	83.21	85.40	85.38	85.06	87.39
	1	823	73.89	63.75	62.32	77.86	78.21	76.68	79.03
	2	266	58.87	44.32	46.31	60.71	62.68	59.13	66.80
	3	58	31.15	15.62	8.06	35.94	36.70	32.55	38.81
	4	16	22.73	10.53	9.30	24.59	27.27	23.33	29.27
	≥ 5	23	40.00	11.11	0.00	43.00	44.55	40.65	46.63

Table 6

Ablation experiment results. Δ denotes the performance drops (F1 Score) under different experimental conditions compared to our proposed model. SA denotes Self-adaptive mask. All the reported results are evaluated using the RoBERTa-base pre-trained language model.

Settings	ACE04		ACE05		GENIA	
	F1	Δ	F1	Δ	F1	Δ
SRT	88.45		88.94		81.87	
w/o BSRM	87.43	−1.02	87.79	−1.15	80.93	−0.94
w/o BTM	87.78	−0.67	88.21	−0.73	81.41	−0.46
w/o BSRM, BTM	87.09	−1.36	87.13	−1.81	80.78	−1.09
w/o SA mask	87.53	−0.92	88.32	−0.62	81.32	−0.55

4.3. Performance on high-degree nested entity

To demonstrate our model's capability in addressing the token overlap challenge in nested named entity recognition (NER), we define the overlap level of an entity based on the number of other entities within which it is nested. For our analysis, entities across our datasets are classified into six distinct levels of nesting, starting from Level 0, which indicates no nesting. For instance, as shown in Fig. 1, the entities "Alabama" and "Florida" are both part of the nested structure "the Alabama-Florida line", which is further nested within "Geneva, near the Alabama-Florida line". This scenario is classified as Level 2 of nesting, indicating two layers of nested entities.

We conducted a comparative analysis using three state-of-the-art models as baselines: PromptNER, DiffusionNER, and CNNNER, each utilizing RoBERTa-base as the pre-trained language model. This analysis was performed on the ACE04 and ACE05 benchmark datasets. As shown in Table 5, SRT underscores the exceptional performance across various levels of entity nesting, with absolute F1 improvements ranging from 2 % to 9 % on both datasets. Whereas baseline performance typically deteriorates as nesting depth increases, SRT's F1 curves remain resilient, demonstrating its ability to capture fine-grained semantic differences and maintain boundary sensitivity in the most challenging nested scenarios.

4.4. Ablation studies

Table 6 presents an ablation study analyzing the performance of the Semantic Refinement and Trimming (SRT) model when key components are selectively disabled. The purpose of this study is to investigate the impact of each module on the overall performance of the SRT model, measured by the F1 score across the ACE04, ACE05, and GENIA datasets. Specifically, the study evaluates the effects of disabling the Boundary-aware Semantic Refinement Module (BSRM), the Boundary Trimming Module (BTM), and the self-adaptive mask (SA mask), either individu-

ally or in combination. The results demonstrate clear performance drops when any of these components are removed, underscoring their critical roles in maintaining the model's ability to accurately detect and classify nested entities. We provide detailed analysis as follows:

Impact of BSRM, BSRM plays a central role in resolving boundary ambiguity by refining the semantic representation of entity boundaries, particularly in cases where tokens are shared between multiple entities. When BSRM is disabled (w/o BSRM), the F1 scores drop by 1.15 % on ACE05 and 0.94 % on GENIA, which highlights the module's importance in distinguishing overlapping tokens that belong to different entities. Without this refinement, the model struggles to correctly identify and classify these entities, leading to boundary errors.

Impact of BTM, BTM serves to eliminate noise by trimming irrelevant or non-entity tokens from the entity boundary predictions. The results show that disabling BTM (w/o BTM) leads to F1-score drops of 0.73 % on ACE05 and 0.46 % on GENIA, indicating that trimming non-entity spans is crucial for improving entity recognition accuracy. This trimming process is particularly important in nested NER, where slight boundary mis-classifications can lead to significant performance losses due to the complexity of overlapping entities.

Combined Impact of BSRM and BTM, Disabling both BSRM and BTM together (w/o BSRM, BTM) yields the most significant performance drops across all datasets, with F1-score declines of 1.36 % on ACE04, 1.81 % on ACE05, and 1.09 % on GENIA. This demonstrates that the synergistic relationship between the two modules is essential for the SRT model's ability to manage complex nested structures. Without these two modules, the model's ability to detect and classify nested entities is significantly compromised, particularly in datasets with higher frequencies of nested entities, such as ACE04 and ACE05.

Impact of Self-adaptive Mask, SA mask is a mechanism that allows the model to dynamically adjust its focus on different parts of the input sequence based on the complexity of the nested entities. Disabling this component (w/o SA mask) leads to moderate performance drops of 0.62 % on ACE05 and 0.55 % on GENIA, indicating its importance in improving SRT's adaptability to diverse entity structures.

4.5. Effect on long entities

In the field of NER, the task of identifying longer entities introduces significant challenges, mainly due to their propensity to encompass nested structures, which exacerbates boundary sensitivity issues. Furthermore, the accurate recognition of these entities represents a "long-tail" challenge, adding complexity to their detection (Wan et al., 2022). In our analysis, we categorize named entities in the ACE04 and ACE05 datasets into six length classes: [1, 4), [4, 7), [7, 10), [10, 13), [13, 16), and [16, +∞). Statistical findings reveal that entities within the [1, 4) length range comprise over 80 % of the dataset entities (81.03 % in ACE04 and 87.58 % in ACE05), while entities longer than 13 characters

Table 7

Performance breakdown on different entity lengths on ACE04 and ACE05. Entities are divided into six groups based on their lengths. "[l]" indicates that the lower bound is included, and "]" means the upper bound is excluded. All models utilized RoberTa-base as the encoder.

Datasets	Entity Length	Proportion in Dataset	DiffusionNER	PromptNER	SRT			
					w/o BSRM	w/o BTM	w/o BSRM, BTM	–
ACE04	[1, 4)	81.03 %	87.17	87.18	89.52	89.87	89.16	90.34
	[4, 7)	11.03 %	53.78	50.68	84.02	84.31	83.84	84.92
	[7, 10)	4.15 %	38.55	37.84	80.90	81.21	80.58	84.37
	[10, 13)	1.51 %	14.89	14.58	70.17	70.51	69.87	73.91
	[13, 16)	0.72 %	16.00	15.09	72.26	72.27	72.14	72.34
	[16, +∞)	1.55 %	13.64	13.06	62.77	63.87	62.10	66.67
ACE05	[1, 4)	87.58 %	86.70	86.35	88.99	89.37	88.31	89.92
	[4, 7)	7.55 %	50.95	50.86	85.50	85.71	85.06	86.04
	[7, 10)	2.32 %	39.36	38.74	74.25	74.48	73.91	80.68
	[10, 13)	1.00 %	32.99	32.99	76.92	77.78	76.27	82.22
	[13, 16)	0.55 %	4.76	7.69	74.26	75.10	73.47	80.00
	[16, +∞)	1.00 %	10.96	8.94	61.11	66.67	60.44	67.50

represent around 2 % in both datasets. This distribution underscores the prevalence of shorter entities and highlights the rarity and complexity of longer named entities within these datasets.

To evaluate the effectiveness of the SRT model in addressing the challenges of long entity recognition, we performed a granular analysis stratified by entity length on the ACE04 and ACE05 benchmarks, comparing against two recent state-of-the-art models: PromptNER (Shen et al., 2023b) and DiffusionNER (Shen et al., 2023a). As illustrated in Table 7, the results indicate that SRT not only performs consistently well across all entity lengths but also significantly outperforms the other models, particularly as the length of entities increases. Notably, while the baseline models demonstrated a decrease in performance with longer entity spans, a common challenge in nested NER due to increasing complexity and boundary ambiguity, SRT maintained consistent and superior performance. As the entity length increases, the performance of the baseline models declines sharply. For example, in the [13, 16) range on ACE05, PromptNER and DiffusionNER achieve F1 scores of only 7.69 % and 4.76 %, respectively, compared to SRT's 80.00 %. A similar trend is observed in ACE04, where PromptNER and DiffusionNER fall short as compared to SRT and its variants.

We further isolate the contributions of our two key modules, Boundary-aware Semantic Refinement Module (BSRM) and Boundary Trimming Module (BTM) via ablation. Removing either module causes a marked drop in F1 for long spans, underscoring their complementary roles: BSRM sharpens semantic distinctions at overlapping boundaries, while BTM prunes spurious, low-quality candidates that disproportionately accumulate for lengthy spans. The fully integrated SRT, combining both modules, therefore achieves uniformly superior performance across all length categories, validating the essential synergy between semantic refinement and boundary trimming in overcoming the long-entity challenge.

4.6. Performance breakdown

Table 8 presents per-entity-type Precision, Recall, and F1 on the ACE04 and ACE05 benchmarks for three leading span-based models (CNNNER, DiffusionNER, PromptNER) alongside our SRT framework. Across both datasets, SRT achieves the highest F1 on the majority of entity categories, six out of seven on both ACE04 and ACE05 datasets, demonstrating broad applicability. In the ACE04 dataset, SRT achieves particularly notable F1 scores in recognizing PER and VEH categories, surpassing other models with scores of 92.45 % and 91.43 %, respectively. In the ACE05 dataset, similar trends are observed with SRT maintaining strong performance, especially in the GPE and WEA categories, achieving F1 scores of 87.90 % and 90.20 %, respectively. Comparatively, the other models show variability in their performance across categories

and datasets. For example, DiffusionNER tends to underperform in categories like FAC and WEA across both datasets.

4.7. Comparison with large language models

Large language models (LLMs), particularly those from OpenAI's GPT family, such as GPT-4 and GPT-3.5, have demonstrated remarkable effectiveness in various natural language processing (NLP) tasks. To assess the efficacy of our proposed SRT model in comparison to state-of-the-art LLMs for information extraction, we conducted a comprehensive evaluation under different settings, including zero-shot, few-shot in-context learning (ICL) (Brown et al., 2020), and few-shot chain-of-thought (CoT) prompting (Wei et al., 2022). The models evaluated include GPT-4, GPT-3.5, LLaMA-3-70B,⁵ PAdELLM-NER, and DEEPSTRUCT, which was derived by applying structured pre-training to a pre-trained 10-billion parameter model. (GLM⁶ Experimental results are illustrated in Table 9, The key findings are:

Specialization vs. Generalization. SRT, which is tailored specifically for nested NER, consistently shows better performance compared to general-purpose LLMs, even when those LLMs are provided with task-specific tuning. On both ACE04 and ACE05 datasets, which are well-known benchmarks for NER, the SRT model achieves 88.45 % (ACE04) and 88.94 % (ACE05). These scores are higher than those of any LLM in any setting. Even the best-performing LLM in the table, GPT-4.0 with 5-shot + CoT, only reaches 46.03 % (ACE04) and 41.46 % (ACE05), which are significantly lower than SRT's scores. This demonstrates that even with task-specific tuning (through few-shot learning with Chain-of-Thought (CoT) prompting), GPT-4.0 underperforms compared to SRT on these NER benchmarks.

Consistency Across Datasets. SRT model shows consistently high performance across all datasets, including 81.87 on Genia and 93.72 on CoNLL03, outperforming all LLMs across the board. In contrast, general-purpose LLMs exhibit more fluctuation in performance across different datasets. For example, GPT-4.0 performs reasonably well in CoNLL03 (76.19 %), but its score drops drastically in ACE05 (41.46 %), highlighting the instability of general-purpose models across different NER tasks compared to SRT's reliable performance.

Impact of Few-shot and In-context Learning. LLMs like GPT-3.5 and GPT-4.0 show marked improvement when moving from zero-shot to few-shot settings, such as GPT-3.5's jump from 27.80 % (zero-shot) to 40.57 % (5-shot + CoT) on ACE04. However, despite this increase, SRT outperforms GPT-3.5 by nearly 48 points in the same dataset, indicating

⁵ <https://huggingface.co/meta-llama/Meta-Llama-3-70B>

⁶ <https://huggingface.co/THUDM/glm-10b>

Table 8
Test performance breakdowns on different entity classes across ACE04 and ACE05.

Settings	Entity Type	CNNNER			DiffusionNER			PromptNER			SRT		
		P	R	F1	P	R	F1	P	R	F1	P	R	F1
ACE04	FAC	70.43	72.32	71.37	67.06	50.89	57.87	57.43	51.79	54.46	84.15	61.61	71.13
	GPE	87.59	88.19	87.89	82.20	80.83	81.51	80.77	81.11	80.94	87.32	91.81	89.51
	LOC	72.12	71.43	71.77	60.17	67.62	63.68	63.72	68.57	66.06	67.19	81.90	73.82
	ORG	79.79	83.70	81.70	72.16	76.09	74.07	72.84	74.82	73.82	82.12	85.69	83.87
	PER	91.22	90.85	91.04	81.64	81.91	81.77	80.52	81.38	80.94	92.98	91.90	92.45
	VEH	84.21	94.12	88.89	75.00	88.24	81.08	70.00	82.35	75.68	88.89	94.12	91.43
ACE05	WEA	60.53	71.88	65.71	66.67	56.25	61.02	67.86	59.38	63.33	87.5	65.62	75.00
	FAC	74.66	80.15	77.30	64.71	68.52	64.71	60.93	67.65	64.11	78.47	83.09	80.71
	GPE	87.01	86.21	86.61	77.35	77.70	77.52	76.63	78.39	77.50	87.30	88.51	87.90
	LOC	73.08	70.37	71.70	61.67	64.71	64.91	58.62	62.96	60.71	65.08	75.93	70.09
	ORG	84.59	81.30	82.91	78.98	77.22	78.09	76.46	77.59	77.02	83.61	84.07	83.84
	PER	89.21	90.91	90.05	81.00	85.86	83.36	81.05	85.47	83.20	91.93	92.65	92.29
	VEH	82.56	69.61	75.53	59.57	54.90	57.14	57.14	54.90	56.00	82.61	74.51	78.35
	WEA	83.33	90.00	86.54	63.79	74.00	68.52	64.91	74.00	69.16	88.46	92.00	90.20

Table 9
Performance comparison with large language models.

Models	Settings	ACE04	ACE05	Genia	CoNLL03
GPT-3.5	Zero-shot	27.80	23.38	38.09	60.10
	5-shot + ICL	38.52	36.17	48.82	70.53
	5-shot + COT	40.57	33.98	50.89	74.63
GPT-4.0	Zero-shot	31.43	24.68	46.22	72.30
	5-shot + ICL	43.35	44.68	56.97	78.50
	5-shot + COT	46.03	41.46	54.47	76.19
LLaMA-3-70B	Zero-shot	17.06	14.35	–	52.47
	5-shot + ICL	36.70	37.10	–	67.24
	10-shot + ICL	43.35	43.50	–	71.01
DEEPSTRUCT (2022a)	Zero-shot	–	28.10	47.20	44.40
PaDeLLM-NER (2024)	Fine-tuning	–	86.90	80.80	93.00
	Instruction Tuning	–	85.02	77.66	92.52
SRT	–	88.45	88.94	81.87	93.72

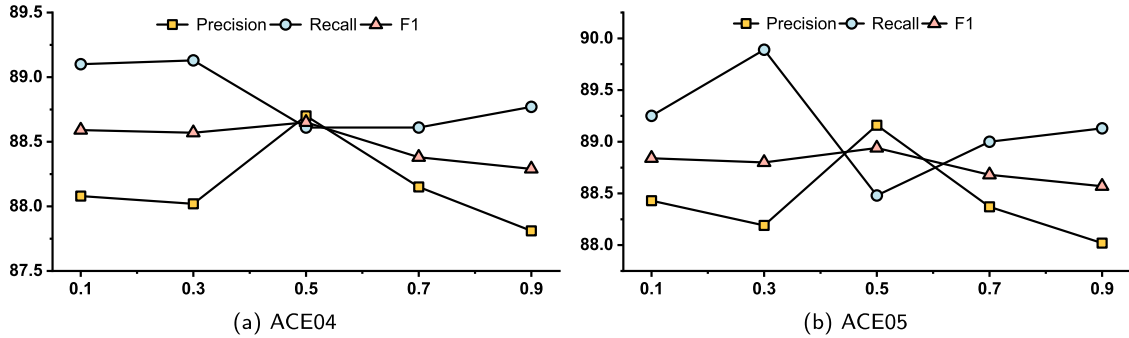


Fig. 4. Sensitivity analysis of the temperature coefficient τ on ACE04 and ACE05 datasets.

that while task-specific tuning improves general-purpose models, it still cannot bridge the performance gap with a specialized model like SRT.

Performance Gap Analysis. LLMs like GPT-3.5 and GPT-4.0 often display oversensitivity to irrelevant context due to their generalized design (Wang et al., 2023). Han et al. (2023) noted that LLMs can become oversensitive to background noise and context that is not directly related to the task, which reduces their accuracy in tasks requiring fine-grained entity detection. In contrast, SRT is specifically designed to focus on boundary dynamics and semantic refinement, enabling it to filter out irrelevant contextual signals, improving its performance in nested NER.

4.8. Stability test for gumbel softmax operation

The Gumbel-Softmax reparameterization is a pivotal technique for facilitating gradient back-propagation through discrete variables, offer-

ing a differentiable approximation of the Argmax function. This is essential in maintaining the differentiability of the BSR operator in non-differentiable scenarios. The Gumbel-Softmax operation employs a formula:

$$z_i = \frac{\exp((\log(\pi_i) + g_i)/\tau)}{\sum_{j=1}^K \exp((\log(\pi_j) + g_j)/\tau)} \quad (12)$$

where, z_i represents the output for category i , g_i is a noise term sampled from the Gumbel distribution, π_i is the log probability of the original input, and K is the total number of categories. This configuration allows the Gumbel-Softmax to approximate discrete distributions more smoothly (Jang et al., 2016) by adjusting the temperature coefficient τ . A lower τ approaches one-hot encoding with higher gradient variances, whereas a higher τ results in smoother distributions and reduced gradient variances. As noted by Gu et al. (2018), the potential training

Table 10

Comparison of runtime (seconds) among various model configurations and baseline models. Δ the difference in inference time relative to our proposed SRT.

Settings/Baselines	ACE04			ACE05		
	Inference Time	Δ	F1	Inference Time	Δ	F1
SRT	49	0	88.45	57	0	88.94
w/o BSRM	47	-2	87.43 (1.02↓)	53	-4	87.79 (1.15↓)
w/o BTM	46	-3	87.78 (0.67↓)	55	-2	88.21 (0.73↓)
w/o BTM and BSRM	42	-7	87.09 (1.36↓)	49	-8	87.13 (1.81↓)
DiffusionNER	193	144	87.42 (1.03↓)	253	196	86.70 (2.24↓)
PromptNER	172	123	87.52 (0.93↓)	215	158	87.05 (1.89↓)

stability issues may arise from the choice of τ during training within the Gumbel-Softmax operation. To empirically test the training stability of SRT, we carefully conducted additional sensitivity analysis on the temperature coefficient τ ranging from 0.1 to 0.9 with increments of 0.2. Experimental results are shown in Fig. 4, several key observations emerge:

1. **Stability across different choices of τ .** Across all test τ values, SRT's performance remains remarkably stable, with F1 fluctuating by less than 0.5 points on both datasets. This insensitivity underscores the resilience of our Boundary-aware Semantic Refinement Module (BSRM) when guided by Gumbel-Softmax sampling.
2. **Optimal temperature around $\tau = 0.5$.** On ACE04 illustrated in Fig. 4, *precision* gently declines from 88.08% at $\tau = 0.1$ to 87.81% at $\tau = 0.9$, whereas *Recall* peaks around 89.1% at $\tau = 0.3$ before converging to 88.7% at $\tau = 0.9$. The highest F1 score is attained at $\tau = 0.5$, striking a balance between boundary sharpness and gradient smoothness. A similar pattern holds on ACE05, where Precision and Recall cross at $\tau = 0.5$, yielding the maximal F1 of 88.94%.
3. **Role of LayerNorm for Gradient Calibration.** We attribute this resilience partly to the LayerNorm transformation immediately following the Gumbel-Softmax operation. By normalizing and rescaling the perturbed logits, LayerNorm mitigates gradient variance and maintains a well-conditioned optimization landscape throughout training (Xu et al., 2019).

4.9. Inference efficiency

To assess the computational efficiency of SRT, particularly the impact of the Boundary-aware Semantic Refinement Module (BSRM) and the Boundary Trimming Module (BTM) on inference times, we performed a detailed comparative analysis. This study utilized the ACE04 and ACE05 datasets and compared SRT's inference efficiency against variations of SRT and state-of-the-art models PromptNER and DiffusionNER. Our experiments were conducted on an Nvidia A100 80G GPU with a batch size of 1. As shown in Table 10, our analysis revealed that integrating BSRM and BTM resulted in a marginal increase in inference time, under 10 seconds. When these modules were activated individually, the additional computational overhead was less than 5 seconds. However, removing both modules led to a performance decrease of 1.36% on ACE04 and 1.81% on ACE05. Importantly, despite these additions, SRT displayed a substantial improvement in time efficiency over the comparative models. Specifically, SRT was 3.51 to 3.93 times faster than PromptNER and DiffusionNER on the ACE04 dataset and 3.77 to 4.43 times faster on the ACE05 dataset. These results underscore SRT's capability to balance enhanced functionality with computational efficiency, maintaining superior performance even with the integration of sophisticated modules.

4.10. Sensitivity test on hyper-parameters

Fig. 5 presents a sensitivity analysis examining the impact of various hyper-parameters, including dropout rate, biaffine size, and CNN

channel dimension on SRT's performance evaluated on ACE04 and ACE05 datasets. As depicted in Fig. 5(a), the optimal dropout rate for both datasets is 0.2, where model performance peaks noticeably. Deviations from this optimal rate in either direction clearly result in performance degradation, suggesting the importance of moderate regularization to balance between preventing overfitting and maintaining adequate model expressivity. Fig. 5(b) illustrates sensitivity to the biaffine attention size, with an optimal setting around 120 dimensions, yielding the highest performance for both datasets. Notably, larger or smaller biaffine sizes lead to declines in performance, likely due to insufficient representational capacity in smaller sizes and potential overfitting or inefficiency in larger configurations. Fig. 5(c) further demonstrates the effect of varying CNN channel dimensions, similarly revealing an optimal dimension around 120, beyond which performance gradually declines. Collectively, this analysis highlights that the model's performance consistently benefits from intermediate hyper-parameter values, suggesting a balanced configuration enhances semantic representational capacity and model generalization.

5. Case studies

Table 11 illustrates detailed prediction outcomes from SRT alongside its variants and SOTA models such as DiffusionNER and PromptNER, as well as GPT-3.5 under the zero-shot setting. The first case features a particularly complex example sentence: "a home of a senior member of the Mukhabarat, Saddam's dreaded secret police", which is characterized by its high nesting level. In this case, SRT distinctively outperforms the SOTA models by accurately identifying all associated nested long entities ending with the same token, including "a home of a senior member of the Mukhabarat, Saddam's dreaded secret police", "a senior member of the Mukhabarat, Saddam's dreaded secret right", and "Saddam's dreaded secret police". The efficacy of SRT in this context is attributed to its integration of the Boundary-aware Semantic Refinement Module (BSRM) and the Boundary Trimming Module (BTM), which are engineered to detect subtle semantic variations and accurately delineate nested entity boundaries, as the elimination of both modules also leads to missing entity identification. The same situation applies to the second case, as both SOTA baselines fail to capture highly nested long entities, including "Donald Zook (ph), who's head of a medical malpractice insurance company" and "head of a medical malpractice insurance company". Meanwhile, SRT exhibits superior performance in identifying boundaries for those nested entities. For the third case, DiffusionNER inaccurately recognizes "neighboring Malaysia" as simply "Malaysia". We doubt this error stems from the certain degree of uncertainty introduced during training, resulting in ambiguous predictions (Du & Li, 2024; Kou et al., 2023) around the ground-truth boundary indices. Additionally, both DiffusionNER and PromptNER fail to classify "F/A-18" as a vehicle, reinforcing their challenges with spanning subtle semantic distinctions.

GPT-3.5 Zero-shot, despite its general-purpose capability in handling various NLP tasks, struggles with highly nested and domain-specific entities in these cases. For instance, GPT-3.5 fails to accurately

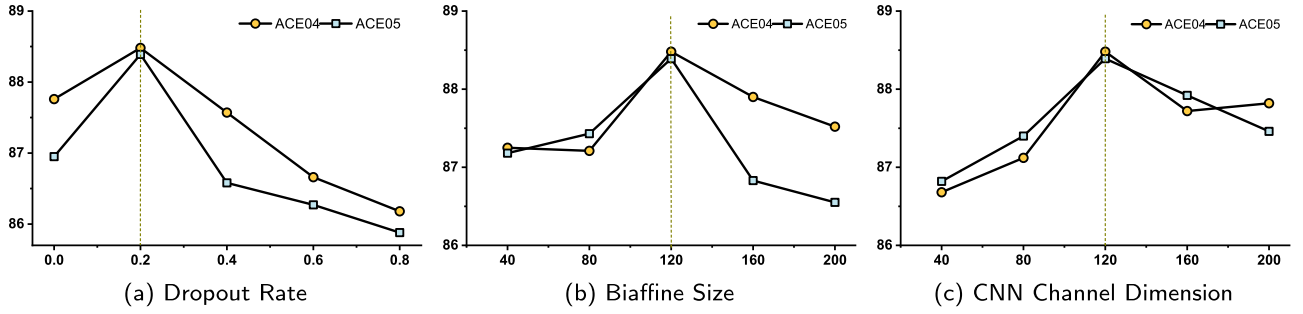


Fig. 5. Sensitivity analysis on different hyper-parameters on ACE04 and ACE05 datasets.

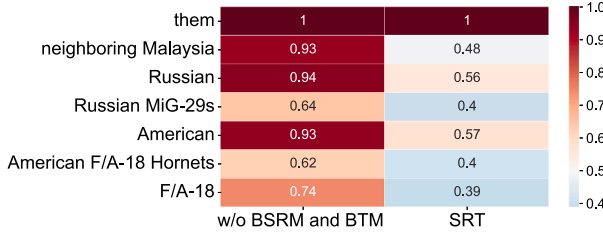
Table 11

Case Study on ACE05. The labels in the lower right corner indicate the entity type, while the superscripts indicate the nesting level. [The candidate entity]_T denotes predicted with incorrect type T. Entities that were missed but should have been identified are denoted in red as {^{m1}The candidate entity^{m1}}, where ^{m1} indicates the nested level of the missed entities. Errors made by baseline models are excluded for brevity.

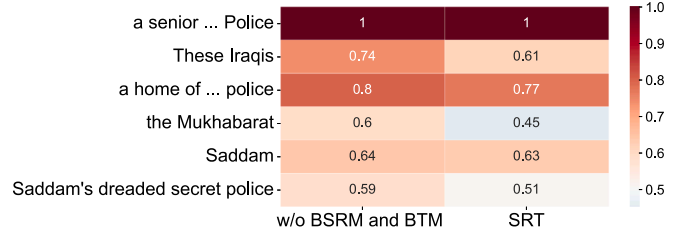
Case 1	
Ground Truth / SRT	[¹ These Iraqis ¹] _{PER} were rifling [¹ a home of [² a senior member of [³ the Mukhabarat ³] _{ORG} , [³ [⁴ Saddam ⁴] _{PER} 's dreaded secret police ³] _{ORG}] ²] _{PER}] ¹ _{FAC} .
w/o BSRM and BTM	[¹ These Iraqis ¹] _{PER} were rifling [¹ a home of [² a senior member of [¹ the Mukhabarat ¹] _{ORG} , { ^{m3} [⁴ Saddam ⁴] _{PER} 's dreaded secret police ^{m3}] ²] _{PER}] ¹ _{FAC} .
DiffusionNER	[¹ These Iraqis ¹] _{PER} were rifling { ^{m1} a home of { ^{m2} a senior member of [¹ the Mukhabarat ¹] _{ORG} , [³ [⁴ Saddam ⁴] _{PER} 's dreaded secret police ³] _{ORG} ^{m2} } ^{m1} }.}
PromptNER	[¹ These Iraqis ¹] _{PER} were rifling { ^{m1} a home of { ^{m2} a senior member of [¹ the Mukhabarat ¹] _{ORG} , [³ [⁴ Saddam ⁴] _{PER} 's dreaded secret police ³] _{ORG} ^{m2} } ^{m1} }.}
GPT-3.5 Zero-shot	{ ^{m1} These Iraqis ^{m1} } were rifling { ^{m1} a home of { ^{m2} a senior member of { ^{m1} the Mukhabarat ^{m1} }, [³ [⁴ Saddam ⁴] _{PER} 's dreaded secret police ³] _{ORG} ^{m2} } ^{m1} }.}
Case 2	
Ground Truth / SRT	[¹ I ¹] _{PER} am concerned about [¹ Donald Zook (ph), [² who ²] _{PER} 's [² head of [³ a medical malpractice insurance company ³] _{ORG}] ²] _{PER}] ¹ ...
w/o BSRM and BTM	[¹ I ¹] _{PER} am concerned about { ^{m1} Donald Zook (ph), [² who ²] _{PER} 's [² head of [³ a medical malpractice insurance company ³] _{ORG}] ²] _{PER}] ^{m1} }.}
DiffusionNER	[¹ I ¹] _{PER} am concerned about { ^{m1} Donald Zook (ph), [² who ²] _{PER} 's { ^{m2} head of [³ a medical malpractice insurance company ³] _{ORG} ^{m2} } ^{m1} }.}
PromptNER	[¹ I ¹] _{PER} am concerned about { ^{m1} Donald Zook (ph), [² who ²] _{PER} 's { ^{m2} head of [³ a medical malpractice insurance company ³] _{ORG} ^{m2} } ^{m1} }.}
GPT-3.5 Zero-shot	{ ^{m1} I ^{m1} } am concerned about { ^{m1} Donald Zook (ph), { ^{m2} who ^{m2} } _{PER} 's { ^{m2} head of [³ a medical malpractice insurance company ³] _{ORG} ^{m2} } ^{m1} }.}
Case 3	
Ground Truth / SRT	But [¹ neighboring Malaysia ¹] _{GPE} 's success in integrating [¹ [² Russian ²] _{GPE} MiG-29s ¹] _{VEH} and [¹ [² American ²] _{GPE} { ^{m2} F/A-18 ^{m2} } _{VEH} Hornets ¹] _{VEH} persuaded [¹ them ¹] _{PER} otherwise...
w/o BSRM and BTM	But [¹ neighboring Malaysia ¹] _{GPE} 's success in integrating [¹ [² Russian ²] _{GPE} MiG-29s ¹] _{VEH} and [¹ [² American ²] _{GPE} { ^{m2} F/A-18 ^{m2} } _{VEH} Hornets ¹] _{VEH} persuaded [¹ them ¹] _{GPE} otherwise...
DiffusionNER	But { ^{m1} neighboring Malaysia ^{m1} } _{GPE} 's success in integrating [¹ [² Russian ²] _{GPE} MiG-29s ¹] _{VEH} and [¹ [² American ²] _{GPE} { ^{m2} F/A-18 ^{m2} } _{VEH} Hornets ¹] _{VEH} persuaded [¹ them ¹] _{PER} otherwise...
PromptNER	But [¹ neighboring Malaysia ¹] _{GPE} 's success in integrating [¹ [² Russian ²] _{GPE} MiG-29s ¹] _{VEH} and [¹ [² American ²] _{GPE} { ^{m2} F/A-18 ^{m2} } _{VEH} Hornets ¹] _{VEH} persuaded [¹ them ¹] _{PER} otherwise...
GPT-3.5 Zero-shot	But { ^{m1} neighboring Malaysia ^{m1} } _{GPE} 's success in integrating [¹ [² Russian ²] _{GPE} MiG-29s ¹] _{VEH} and [¹ [² American ²] _{GPE} { ^{m2} F/A-18 ^{m2} } _{VEH} Hornets ¹] _{VEH} persuaded { ^{m1} them ^{m1} } otherwise...

detect nested entities such as “the Mukhabarat”, “a senior member of the Mukhabarat, Saddam’s dreaded secret police” in the first case. This underscores the limitations of general-purpose models like GPT-3.5 in specialized tasks such as nested NER, where task-specific modules such as BSRM and BTM play a critical role in improving entity boundary precision.

To better illustrate the effectiveness of our proposed BSRM and BTM, we provide semantic similarity heatmaps for two cases selected from Table 11. As shown in Fig. 6, SRT effectively models the semantic differences between spans, mitigating the issue of boundary insensitivity, especially in cases with overlapping or nested entities.



(a) Semantic Similarity Visualization of Case 1



(b) Semantic Similarity Visualization of Case 3

Fig. 6. Semantic similarity visualization for all named entities in Case 1 and 3. We compare SRT with its variant: SRT w/o BSRM and BTM. For both cases, we select the most likely misclassified entities: “them” and “a senior member of the Mukhabarat, Saddam’s dreaded secret police” and compute the semantic similarity heatmaps between these two misclassified entities and the remaining named entities in each case.

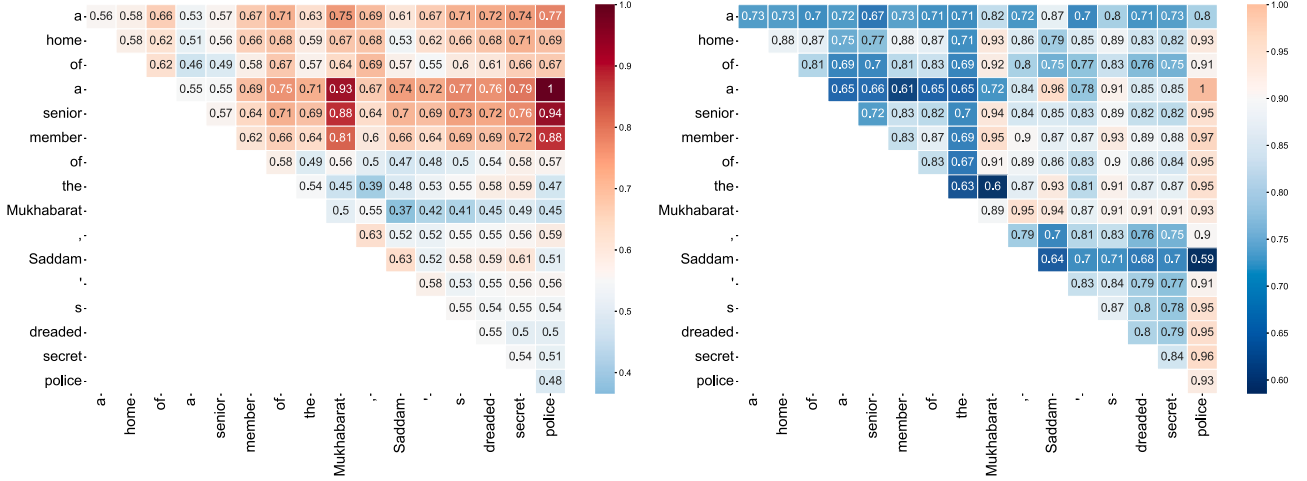


Fig. 7. Semantic similarity visualization between entities and non-entities in Case 3. We compare SRT with its variant: SRT w/o BSRM and BTM. In this case, we select the misclassified entity “a senior member of the Mukhabarat, Saddam’s dreaded secret police” and compute semantic similarity heatmaps between this entity and the remaining spans in the sentence. For brevity, we focus on the highly nested portion of the sentence, highlighting the impact of BSRM and BTM in addressing boundary sensitivity.

Table 12

Error analysis on the ACE04 dataset. Missing predictions are **highlighted**, ground-truth entities are underlined, and the correct entity type is denoted within $[\text{PER}]$. Correctly predicted entities are omitted for brevity.

Context:	<u>It</u> _[PER] ’s the tip of the tip of the spear _[PER] .
SRT	It’s the tip of the tip of the spear _[PER] .
DiffusionNER	It’s the tip of the tip of the spear _[PER] .
PromptNER	It’s the tip of the tip of the spear _[PER] .
Context:	<u>It</u> _[FAC] lies just 12 miles from <u>Baghdad</u> _[GPE] and will be a key forward base for <u>U.S. troops</u> _[PER] <u>as they</u> _[PER] prepare for a push on the <u>capital</u> _[GPE] .
SRT	It lies just 12 miles from Baghdad and will be a key forward base for <u>U.S. troops</u> _[FAC] as they prepare for a push on the capital.
DiffusionNER	<u>It</u> _[FAC] lies just 12 miles from Baghdad and will be a key forward base for <u>U.S. troops</u> _[FAC] as they prepare for a push on the capital.
PromptNER	<u>It</u> _[FAC] lies just 12 miles from Baghdad and will be a key forward base for <u>U.S. troops</u> _[FAC] as <u>they</u> _[PER] prepare for a push on the capital.

In the variant without BSRM and BTM, we observe a relatively high semantic similarity between most entities, with over half displaying a similarity exceeding 70 % when compared to the entity “them”. This indicates that, without these modules, the model struggles to capture fine-grained distinctions between similar entities, leading to potential mis-classifications due to boundary confusion. On the other hand, with both BSRM and BTM integrated, the semantic similarity between entities decreases significantly. This highlights the ability of BSRM and BTM

to refine span-level semantic differences, allowing the model to better distinguish between entities and reduce overlap in semantic representation. A similar trend is observed in other cases, where the inclusion of BSRM and BTM consistently leads to a decrease in semantic similarity between entities.

To illustrate how our proposed BSRM and BTM effectively filter out non-entity spans, we showcase a semantic similarity heatmaps between entities and non-entity spans in Fig. 7. When excluding BSRM and BTM, those non-entity spans ending with the token “police” (the last column of the heatmaps) holds relatively high similarities with the truth entity “a senior member of the Mukhabarat, Saddam’s dreaded secret police”. This high similarity leads to boundary confusion and misidentification, causing the variant of SRT without these modules to miss some entities within this highly nested portion of the sentence (“Saddam’s dreaded secret police”). The inclusion of BSRM and BTM is crucial in reducing such overlap, allowing the model to better distinguish between entities and non-entities, thereby improving boundary detection.

6. Error analysis

In this section, we provide a detailed error analysis comparing our proposed SRT method with two state-of-the-art baselines: DiffusionNER and PromptNER. As shown in Table 12, two representative cases from the ACE04 dataset are analyzed to illustrate model performance, particularly focusing on the challenges posed by long entity spans and intricate nested entity structures. In the first example, the context presents significant linguistic complexity due to repetitive tokens: “It_[PER]’s the tip of the tip of the spear_[PER]”. Here, the correctly

annotated entity span, “the tip of the tip of the spear,” comprises eight tokens, posing notable difficulties for span-based entity recognition models. Despite their advanced architectures, all three methods fail to correctly identify this elongated entity span. Such consistent mis-classification underscores a common shortcoming across contemporary NER approaches in reliably capturing semantic boundaries within lengthy, which was previously shown poor performance of long entity lengths in Table 7. The second example further amplifies this challenge by introducing complex nested and overlapping entities: the full entity “a key forward base for U.S. troops_[FAC]” (an 11-token entity), which internally contains the embedded entity “U.S. troops_[PER].” Again, due to the considerable span length and embedded structure, all evaluated models fail to correctly extract the full 11-token entity. Nevertheless, it is worth highlighting that, despite this limitation, our SRT approach demonstrates a comparatively stronger performance than the baseline methods. Specifically, both DiffusionNER and PromptNER exhibit notable additional errors, failing to correctly recognize crucial short-span entities (“it” or “they”) within the same sentence. In summary, these illustrative examples indicate a general challenge across current state-of-the-art methods in accurately extracting extended entity spans. Nonetheless, our proposed SRT shows relative advantages in complex nested contexts through improved recognition of smaller entities within overlapping scenarios.

7. Conclusion and future works

In this work, we present Semantic Refinement and Trimming (SRT), a boundary-aware network specifically crafted to resolve the pervasive issue of ambiguous span boundaries in nested Named Entity Recognition. SRT’s effectiveness stems from two dedicated modules: (1) the Boundary-aware Semantic Refinement Module (BSRM), which employs learnable convolutional kernels to detect and amplify subtle semantic divergences among overlapping spans, and (2) the Boundary Trimming Module (BTM), which prunes low-quality, non-entity candidates through a dual-pathway architecture that balances high-level abstraction with fine-grained detail restoration. Extensive experiments on the ACE04, ACE05, CoNll03, and GENIA benchmarks confirm that SRT consistently surpasses state-of-the-art baselines across flat and nested NER tasks. Ablation analyses further establish the indispensable contributions of both BSRM and BTM to overall performance, while qualitative case studies vividly illustrate their complementary roles in refining semantic boundaries. Future work will investigate the adaptability of our SRT framework to discontinuous NER, aiming to extend its boundary-sensitive refinement capabilities to even more complex entity recognition scenarios.

CRediT authorship contribution statement

Yanglei Gan: Writing – original draft, Methodology, Conceptualization; **Yao Liu:** Writing – review & editing, Validation; **Yuxiang Cai:** Visualization, Software; **Run Lin:** Software, Investigation; **Song Yang:** Validation, Supervision; **Qiao Liu:** Supervision, Methodology, Conceptualization; **Yashen Wang:** Validation, Investigation; **Xiaojun Shi:** Validation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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